

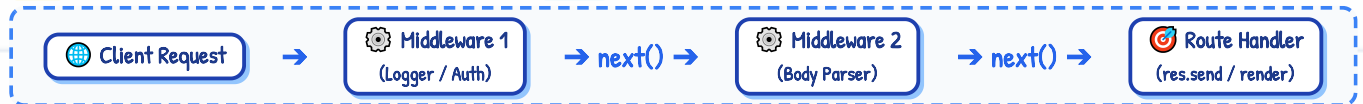


07. Express Middleware Fundamentals



What is Middleware?

Middleware ek aisa function hota hai jo Request receive hone aur Response bhejne ke beech execute hota hai. Iske paas **req**, **res**, aur agle function ko trigger karne ke liye **next** ka access hota hai.



Middleware kya kar sakta hai?

- Koi bhi code execute kar sakta hai (e.g. request timing, analytics).
- Request aur Response objects ko modify kar sakta hai (e.g. `req.user = decodedToken`).
- Request-Response cycle ko end kar sakta hai (e.g. `res.status(401).send("Unauthorized")`).
- Stack me agle middleware ko **next()** ke zariye call kar sakta hai.



Custom Logger Middleware Example

```
index.js - Custom Logger Middleware

app.use((req, res, next) => {
  req.responseTime = new Date(Date.now()).toString();
  console.log(`[${req.method}] ${req.path} at ${req.responseTime}`);
  next(); // 🚨 Important: next() call nahi kiya toh request hang ho jayegi!
});
```



NEVER FORGET next(): Agar aap middleware me na toh response bhej rahe hain aur na `next()` call kar rahe hain, toh browser infinite loading me phas jayega!



Error-Handling Middleware & Custom Errors



Error-Handling Middleware: 4 Arguments Signature

Error handling middleware me hamesha 4 parameters hote hain: **(err, req, res, next)**. Express is signature se pehchanta hai ki ye special error handler hai!

next(): Flow ko agle regular middleware ya route handler ke paas bhejta hai.

next(err): Saare bache hue regular routes ko skip karke seedha Error Middleware par jump karta hai!



Custom Error Class (ExpressError)

Custom HTTP status codes aur structured messages maintain karne ke liye standard class:

```
ExpressError.js - Custom Error Class

class ExpressError extends Error {
  constructor(status, message) {
    super();
    this.status = status;
    this.message = message;
  }
}
module.exports = ExpressError;
```



Global Error Handler & 404 Route

```
index.js - Error Handling Setup

// 1. 404 Not Found Handler (sabse neeche)
app.all("*", (req, res, next) => {
  next(new ExpressError(404, "Page Not Found!"));
});

// 2. Global Error Handling Middleware
app.use((err, req, res, next) => {
  let { status = 500, message = "Something Went Wrong!" } = err;
  res.status(status).render("error.ejs", { message });
});
```